direct duplications. Proteins fail to fulfill this criteria.

- **Stability:** It should be stable structurally and chemically. The genetic material should be stable enough not to change with different stages of life cycle, age or with change in physiology of the organism. Stability as one of property of genetic material was clearly evident in Griffith's transforming principle. Heat which killed the bacteria did not destroy some of the properties of genetic material. In DNA the two strands being complementary, if separated (denatured) by heating can come together (renaturation) when appropriate condition is provided. Further 2' OH group present at every nucleotide in RNA is a reactive group that makes RNA liable and easily degradable. RNA is also known to be catalytic and reactive. Hence, DNA is chemically more stable and chemically less reactive when compared to RNA. Presence of thymine instead of uracil in DNA confers additional stability to DNA.

- **Information storage:** It should be able to express itself in the form of '**Mendelian characters**'. RNA can directly code for protein synthesis and can easily express the characters. DNA, however depends on RNA for synthesis of proteins. Both DNA and RNA can act as a genetic material, but DNA being more stable stores the genetic information and RNA transfers the genetic information.

- **Variation through mutation:** It should be able to mutate. Both DNA and RNA are able to mutate. RNA being unstable, mutates at a faster rate. Thus viruses having RNA genome with shorter life span can mutate and evolve faster.

The above discussion indicates that both RNA and DNA can function as a genetic material. DNA is more stable, and is preferred for storage of genetic information.

## 5.7 Packaging of DNA helix

The distance between two consecutive base pairs is 0.34nm ($0.34\times10^{-9}$m) of the DNA double helix in a typical mammalian cell. When the total number of base pairs is multiplied with the distance between two consecutive base pairs ($6.6 \times 10^{9} \times 0.34 \times10^{-9}$ m/bp), the length of DNA double helix is approximately 2.2 m. (The total length of the

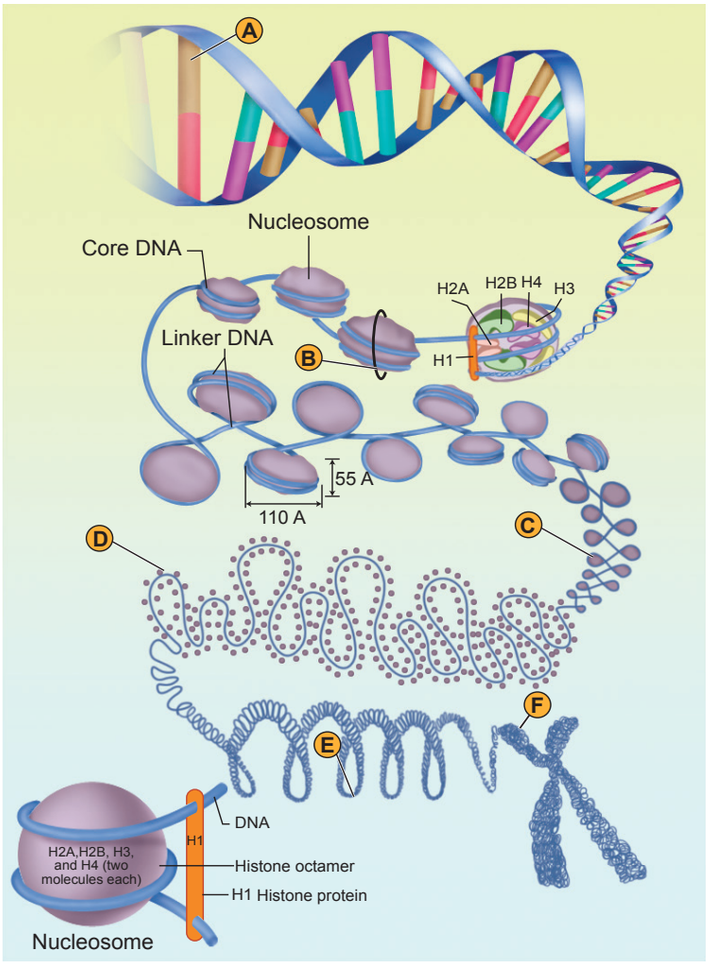



**Fig. 5.3 Condensation of DNA - A - DNA, B-Nucleosomes and Histones, C- Chromatin fiber, D- Coiled chromatin fiber, E- Coiled coil, F- metaphase chromatid**



XII Std Zoology Chapter 5 EM.indd 67 23/12/2022 16:07:12